

Litwell

How-to Guide

Version 1.4.3
27 October 2026

Search papers, preprints and patents in one go,
and get one tidy, ranked Excel file.

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Litwell is free and open source. This guide is written for any researcher; no programming is needed.

1. What Litwell does

Litwell searches the scientific literature, preprints and patents for you, all at once, and saves what it finds to one tidy Excel workbook. You type a topic, tick the places to look, and click Search. Litwell asks every source, removes duplicates, filters out the obvious misses, scores what is left from 0 to 100, and sorts the best matches to the top.

It was built for the common problem of a search that returns far too much: an abbreviation that means several things (B2 can be a viral protein, vitamin B2, or a breast-cancer receptor), a phrase that is split across unrelated words, or a topic spread over papers, preprints and patents. Litwell searches phrases as phrases, lets you say what must and must not appear, and shows you exactly what every source and every filter did.

Litwell is free and open source, runs on your own computer, needs no account, and sends your searches only to the public sources you tick. It is a search tool, not a judge of quality: a high score means a record matches your words well, not that the science is sound.

2. Quick start: your first search in 5 steps

1. Open Litwell. Choose **File, Set Output File** and pick where your workbook should be saved (for example `My Search.xlsx`).
2. Type a **Topic / keyword**, for example `double-stranded RNA`. Leave the match mode on **Exact phrase**.
3. Tick the sources you want. **Papers only** is a good start: it skips the two patent sources, which need free keys (see section 5).
4. Click **Search**. The Status box shows each source as it answers, then a summary: what each source returned, what your filters removed, and the best matches.
5. Click **Open File**. The Results sheet is sorted with the best matches first; read the Summary and Search Log sheets to see what happened.

Search again with a different topic into the same file and Litwell merges the new records in, marks which rows are new, and never duplicates a paper.

> Everything in Litwell is free. The only keys it ever asks for are free ones (USPTO and EPO for patents; NCBI and OpenAlex are optional), and none is needed to search papers and preprints.

3. Every field explained

This section describes every box on the main window, in the order it appears. Each entry has the same layout: what it does, the values it accepts, its default, and when to use or skip it.

Topic / keyword

The words you are looking for. With the default match mode this is searched as one exact phrase in titles and abstracts.

- **Valid values:** Any text. Do not add quotes or AND/OR yourself; use the match mode instead.
- **Default:** Empty
- **Use it when:**
  - You know the subject: a protein, a technique, a disease, a technology.
- **Skip it when:**
  - You only want a particular author or journal; leave it empty and use those boxes.
- **Related:** Match topic as, Must also contain, Exclude terms

Author

Restricts the search to papers by a person (or, for patents, an inventor).

- **Valid values:** A name such as `Smith J` or `Marie Curie`.
- **Default:** Empty
- **Use it when:**
  - You want everything by one researcher, or a topic by one group.
- **Skip it when:**
  - Searching the preprint source on its own: one of its backends (OSF) has no author search and is skipped.
- **Related:** Journal

Title contains

Words that must be in the title, searched as a phrase.

- **Valid values:** Any text.
- **Default:** Empty
- **Use it when:**
  - The topic is common and you want only papers that are really about it, which shows in the title.
- **Skip it when:**
  - You are happy for the words to appear only in the abstract.
- **Related:** Topic / keyword

Journal

Restricts papers to one journal. For patents this is the applicant (the company or institution). In the preprint source it names the server, for example `bioRxiv`.

- **Valid values:** A journal, server or applicant name.
- **Default:** Empty
- **Use it when:**
  - You follow one journal, or want a particular company's patents.
- **Related:** Author

Date range

Limits results to a publication period. Both ends are optional.

- **Valid values:** `YYYY`, `YYYY-MM` or `YYYY-MM-DD`.
- **Default:** Empty (all dates)
- **Use it when:**
  - You want recent work, or a fixed window for a review.
- **Skip it when:**
  - You are not sure when the work appeared; leave it open and sort by date in Excel.

Match topic as

How the topic words must appear. See section 4 for a worked example.

- **Valid values: Exact phrase**, **All words** or **Any word**.
- **Default:** Exact phrase
- **Use it when:**
  - Exact phrase: the words belong together, such as `B2 protein` or `dot blot`.
  - All words: you want every word but in any order or position.
  - Any word: you are brainstorming and will filter afterwards.
- **Related:** Must also contain, Exclude terms

Must also contain (any of)

A comma-separated list. The title or abstract must contain at least one of the terms, or the record is removed. Use it to demand context.

- **Valid values:** Terms separated by commas, for example `protein, viral, RNAi, assay`.
- **Default:** Empty (no requirement)
- **Use it when:**
  - The topic is ambiguous and you know words that real hits will also contain.
- **Skip it when:**
  - Many records have no abstract (some Crossref and all USPTO records); a strict list removes those unless the title has a term.
- **Related:** Exclude terms, Profile

Exclude terms

A comma-separated list of things you do not want. A term in the title or journal name removes the record. A term found only in the abstract keeps the record but lowers its score by 20 points per term, up to 40.

- **Valid values:** Terms separated by commas, for example `riboflavin, vitamin, HER2`.
- **Default:** Empty
- **Use it when:**
  - A second meaning of your topic keeps flooding the results.
- **Skip it when:**
  - The term could also appear in genuinely relevant records; prefer Must also contain, or accept the score penalty.
- **Related:** Must also contain, Relevance Score

What counts as relevant (for AI)

A plain-language description of what you are after. It is used only by the optional AI triage, which judges each record against it, and it is saved with a profile.

- **Valid values:** One or two sentences.
- **Default:** Empty (the AI then uses your search terms)
- **Use it when:**
  - You switched on AI triage.
- **Skip it when:**
  - You are not using AI triage.
- **Related:** AI triage

Profile

A saved set of the fields above plus the ticked sources. Pick one to fill the form, type a new name and click Save to store the current form, or Delete to remove one.

- **Valid values:** A name. Two examples ship with Litwell, both marked `Example:`; delete them if you do not want them.
- **Default:** None
- **Use it when:**
  - You repeat the same kind of search.
- **Related:** Search profiles (section 7)

Max results (per source)

The most records Litwell asks each source for. Filters are applied to what was fetched.

- **Valid values:** 1 to 500 (PubMed 500, most others 200, patents 100).
- **Default:** 100
- **Use it when:**
  - Raise it for a broad search, lower it for a quick look.
- **Skip it when:**
  - With a very loose search a small cap can leave few records after filtering; tighten the search instead.

Hide scores below

Removes records whose Relevance Score is lower than this number. The removal is counted in the Search Log.

- **Valid values:** 0 to 100.
- **Default:** 0 (keep everything, sorted)
- **Use it when:**

    - You want a short list and trust the score.
- **Skip it when:**
    - You are still learning what a good score looks like for your topic.
- **Related:** Relevance Score

AI triage (local Ollama, optional)

Asks a model running on your own computer to rate each record Relevant, Maybe or Not relevant, with a one-line reason. Off by default. See section 11.

- **Valid values:** On or off.
- **Default:** Off
- **Use it when:**
    - You have Ollama installed and want a second opinion on a long list.
- **Skip it when:**
    - You do not have Ollama, or want the fastest possible search.
- **Related:** What counts as relevant

Search in (sources)

The databases to search. Tick any number; they are asked at the same time. **All**, **Papers only** and **None** are shortcuts.

- **Valid values:** PubMed, Crossref, OpenAlex, Europe PMC, arXiv, Preprint servers, USPTO, EPO.
- **Default:** Whatever you ticked last time
- **Use it when:**
    - See section 5 for how to choose.
- **Related:** Free keys

4. Match modes, with a worked example

The match mode decides how the words of your topic must appear. Each source is sent the query in its own native syntax (for example PubMed's `[Title/Abstract]`, OpenAlex's `title_and_abstract.search`, EPO's CQL). Crossref and EPO cannot match a phrase themselves, so Litwell checks their results against your phrase afterwards.

Mode	Means	Example: topic `B2 Protein`
Exact phrase	The words together, in order	finds "the B2 protein of nodaviruses", not "vitamin B2 and a protein"

Mode	Means	Example: topic `B2 Protein`
All words	Every word, anywhere, any order	finds both of the above
Any word	At least one word	finds anything with B2, or with protein

Worked example: B2 Protein

B2 is an ambiguous abbreviation. A search for `B2 Protein` without quotes (how earlier versions searched) returned **1,219 rows** in the test file. Only about 16% of those had B2, beta-2, FHV or dsRNA in the title; the rest were vitamin B2, HER2, heart failure and food science. Only **10** had the phrase B2 Protein in the title.

The same search as an exact phrase, then with the *Example: Ambiguous abbreviation* profile (must contain `protein, viral, virus, RNAi, assay`; exclude `riboflavin, vitamin, HER2`):

Source	Matches before (plain words)	Matches now (exact phrase)
PubMed	57,749	137
OpenAlex	304,231	253
Europe PMC	205,957	146
Preprint servers	10,919	8
EPO	654	33 (13 more removed by the phrase check)
USPTO	not searched	1
Crossref	1,447,717 (ranked by similarity)	200 fetched, only 9 contain the phrase

Run	Rows in the workbook	Rows scoring 50 or more
Earlier version, plain words, all sources	1,219	not scored
Exact phrase, all sources	241	51
Exact phrase with the example profile	217	117

In the profile run, the first eleven rows (scores 95 to 100) are all about the B2 protein of nodaviruses or the Flock House virus. Further down, papers that use "B2 protein" for something else (a wheat protein, a bacterial gene) appear with lower scores, and papers that only mention the words in passing score lowest. A phrase can still be ambiguous, so read the top of the list first and use Must also contain and Exclude terms to sharpen it.

> Search counts change from day to day, and the numbers above come from one test run. Treat them as an illustration of the size of the effect, not as fixed figures.

5. Choosing sources, and getting the free keys

Source	Best for	Needs
PubMed	Biomedical and life-science papers, with good abstracts	nothing
Crossref	Papers in every field; the broadest record of who published what	nothing
OpenAlex	All disciplines; abstracts, citation counts and open-access links	nothing
Europe PMC	Biomedical papers plus preprints; free full-text links	nothing
arXiv	Physics, maths, computer science, quantitative biology	nothing
Preprint servers	bioRxiv, medRxiv, SSRN, ChemRxiv, OSF servers and many more, found through three aggregators	nothing
USPTO	US patents and published applications	free key
EPO	Patents from about 100 offices	free key and secret

Papers, preprints and patents

- **Papers** have been peer reviewed. PubMed, Crossref, OpenAlex and Europe PMC overlap heavily; Litwell merges the overlap into one row and fills in what one copy lacks.
- **Preprints** are shared before peer review. They are the newest work and also the least checked. The Type column says Preprint so you can filter them in or out.
- **Patents** describe inventions rather than findings. In the workbook the inventors are in Authors, the applicant is in Journal, and the patent number is in DOI. An application that is not yet granted is marked `(application)` after its date.

Getting the free keys

Paste keys under **File, API Keys & Contact**. They are saved only on your computer.

- **USPTO:** sign in or create a free account at data.uspto.gov, generate or copy your API key, and paste it into *USPTO API key*.
- **EPO:** register free at developers.epo.org, create an app to get a *Consumer Key* and a *Consumer Secret*, and paste them into *EPO OPS key* and *EPO OPS secret*. The free tier allows about 4 GB a week.
- **NCBI (optional):** a free key from your NCBI account (Settings, API Key Management) raises PubMed's limit from 3 to 10 requests a second. Most people do not need it.
- **OpenAlex (optional):** a free key gives you your own daily allowance if you ever see "rate limit reached".

If USPTO or EPO is ticked but its key is missing, Litwell does not fail the whole search. It warns you in the Status box, skips that source, runs the others, and records *Skipped: no key* in the Search Log.

6. The contact email setting

Crossref, OpenAlex and NCBI run a faster, friendlier lane for programs that identify a contact. If you type an email under **File, API Keys & Contact, Contact email**, Litwell includes it in requests to those three services so they can reach you if something misbehaves.

- It is **optional** and **blank by default**. Litwell works without it.
- It is stored **only** in your own settings file (`.litsearch_config.json` in your home folder) and is never sent anywhere except to those three services.
- It is never built into the app. Each person who uses Litwell adds their own, if they want one.

7. Search profiles and exclusion terms

A **profile** remembers a precise search so you can repeat it: the topic, match mode, must-contain and exclude lists, the plain-language description for AI triage, the minimum score and the ticked sources. Pick a profile from the Profile box to fill the form; type a name and click **Save** to store the current form; **Delete** removes one.

Two examples are included, clearly labelled and safe to delete. They are only illustrations of the kind of thing a profile holds.

Example profile	Topic and rules
Example: Ambiguous abbreviation	Topic `B2 Protein`, exact phrase. Must contain `protein, viral, virus, RNAi, assay`. Exclude `riboflavin, vitamin, HER2`.
Example: Patent landscape	Topic `double-stranded RNA detection`, all words. Must contain `antibody, assay, kit, immunoassay, sensor`. Sources USPTO, EPO and OpenAlex.

Tips for exclusion terms

- Terms match **whole words**, ignoring capitals; `HER2` does not match `HER23`. A space and a hyphen are interchangeable, so `B2 protein` also finds `B2-protein`.
- Start with two or three terms for the second meaning you keep seeing, then look at the Search Log to see how many rows each one removed.
- If a term removes good records too, move it out of *Exclude* and rely on the score: a term found only in the abstract already costs 20 points.

8. How the Relevance Score is calculated

Every record gets a number from 0 to 100 using fixed rules. No AI is involved and no key is needed. The score is a guide to how well the record matches your words; it is not a judgement of quality.

What Litwell looks for	Points
Your topic as a phrase in the **title**	50
If not as a phrase: the share of your topic's words in the title	up to 30
Your topic as a phrase in the **abstract**	25
If not as a phrase: the share of your topic's words in the abstract	up to 15
Each *Must also contain* term found in the title or abstract	10 each, up to 20
The journal matches the Journal box, or a field in the profile	5
Each *Exclude* term found only in the abstract (a title or journal hit removes the record)	minus 20 each, up to 40

The total is kept between 0 and 100. If you did not type a topic or a title (for example an author-only search), there is no text to judge and every record scores 50. The Title box counts as the topic when the Topic box is empty.

An example

Topic `B2 protein`, must contain `viral`, exclude `vitamin`. A paper titled "The B2 protein of a virus" whose abstract mentions a viral B2 protein scores 50 + 25 + 10 = 85. A paper called "Cheese ripening" whose abstract mentions vitamin B2 proteins scores a few points: no title match, a partial abstract match, minus 20 for the excluded word.

Hide scores below removes records under a number you choose. Sorting best first is usually enough; use it for a short list once you know what a good score looks like for your topic.

9. Reading the workbook

Results

One row per paper, best first. Header frozen, filters on every column, abstracts wrapped (click a cell to read the whole abstract).

Column	What it tells you
Relevance Score	0 to 100, coloured from red to green

Column	What it tells you
AI Rating, AI Reason	From the optional local AI triage; blank when it was off
New Since Last Run	Yes if the record was not in the file before the latest run (section 10)
Title, Authors, Journal, Date	As published. For patents: inventors, applicant, publication date
Abstract	When any source supplied one
DOI, Link, Open-access link	Identifier, a page for the record, and a free full-text link when one is known
Citation count	How many works cite it (the highest figure any source reported)
Type	Paper, Preprint or Patent
PubMed ... EPO	Yes or No: which sources returned it. Filter a column to Yes to see one source's records
Search Date, Search Terms	When it was first found and with which fields

Summary

Counts by source, type, year, AI rating and score band, plus how many rows the latest run added and removed. It is rebuilt every run.

Search Log

One row per source per run, including sources that failed or were skipped (with the reason), and one row per filter rule with the number of rows it removed. It answers "why is this missing?" and "why only this many?".

Glossary

Every column and scoring term explained, inside the workbook, with the credit and license at the foot.

You can also save to `.csv`, `.txt`, `.md`, `.ris` or `.bib`. Those files receive the same filtered, scored records but not the extra sheets.

10. New Since Last Run

When the output file already exists, each run marks which rows are new. Search the same topic every month into the same file, and filter *New Since Last Run* to **Yes** to see only what appeared since last time.

- **Yes:** the record was not in the file before this run.
- **No:** it was already there (even if this run found it again). Records found again also gain any extra source flags and missing details.
- **Blank:** the first run into a brand-new file, when there is no earlier run to compare with.

Files made by older versions are upgraded the first time they are searched into: nothing is lost, old rows are marked No, and the extra columns start empty for them.

> Close the workbook in Excel before searching into it. If it is open, Litwell cannot update it; it saves the results to a recovery CSV next to it so nothing is lost.

11. Optional local AI triage (Ollama, free)

AI triage is a second opinion for long lists. It is **off by default**, runs **only on your own computer**, and costs nothing. Litwell never sends records to a cloud AI service, and refuses any Ollama address that is not your own computer.

Setting it up once

1. Install Ollama from ollama.com (free, Windows and Mac).

2. Open a terminal and pull a model, for example: `ollama pull llama3.1`. Any chat model will work, and Ollama may already have one.

3. Nothing to set in Litwell: with the **Ollama model** box (File, API Keys & Contact) left blank, Litwell detects an installed model automatically and tells you which one it used, in the Status box and in the Search Log. To pick a model yourself, or to switch to a newer one later, type its name there; no update of Litwell is needed.

4. Describe what you want in **What counts as relevant**, tick **AI triage**, and search.

For each record Litwell shows the model the title, journal and abstract plus your description, and asks for a rating (**Relevant**, **Maybe**, **Not relevant**) and a one-line reason. These fill the *AI Rating* and *AI Reason* columns. Records are rated best-score first, up to 200 per run (setting `ai_max_records`), and progress shows in the Status box. Expect roughly a second or a few seconds per record on a typical laptop.

- If Ollama is not installed or not running, the Status box shows one line (for example *AI triage skipped: Ollama is not running on this computer*) and the search carries on without it.
- A model can be wrong. Treat the rating as a prompt to look, not a decision, and read the reason.
- Records with no abstract are rated from the title alone and usually come back *Maybe*.

12. Troubleshooting

What you see	What it means and what to do
No results	The exact phrase may be too narrow. Try **All words**, check the spelling, widen the date range, or remove Must also contain. The Search Log shows how many each source matched and how many your rules removed.
A source says **skipped** or **NOT searched**	The reason is shown. *No USPTO key* or *no EPO key*: add the free key (section 5). *Could not reach ... HTTP 503*: that service is down; try again later. The other sources still ran.
Rate limit reached	A service asked you to slow down. Wait and search again. For OpenAlex a free key or a contact email helps (sections 5 and 6).
Missing key warning	USPTO or EPO is ticked without a key. Add it under File, API Keys & Contact, or untick the source.
Could not update ... it is open in Excel	Close the workbook and search again. The results were saved to a recovery CSV next to the file.
Far more rows than expected	Use the exact phrase mode, add Must also contain or Exclude terms, or set Hide scores below.
Far fewer rows than expected	Lower the strictness: All words, fewer must-contain terms, a larger Max results. Crossref and USPTO records often have no abstract, so a must-contain rule can remove them.
The score seems wrong	Read section 8. The score reflects your words, not quality. A paper with the phrase only in the abstract scores lower than one with it in the title.
macOS says the app cannot be opened	The app is not code-signed. Open System Settings, Privacy & Security, and choose Open Anyway.
Windows says Windows protected your PC	Click More info, then Run anyway.
AI triage does nothing	Check that Ollama is running and has a chat model (`ollama list`). If you typed a model name and it is not installed, Litwell uses an installed model instead and says so; the Status box gives any other reason.

13. Credits and license

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Litwell does not collect anything. Your searches go only to the public sources you tick. Each database and patent office has its own terms of use, which you are responsible for following.

14. Glossary

Abstract

The short summary at the start of a paper or patent.

AI Rating

Relevant, Maybe or Not relevant, from an optional local Ollama model judging the title and abstract against the profile description. Blank when AI triage was off or unavailable.

AI Reason

One line from the local model explaining its rating.

All words / Any word

Every word must appear (any order) / at least one word must appear.

Application (patent)

A published patent application that has not (yet) been granted. Marked `(application)` after the date.

Boolean search

Combining words with AND, OR and NOT. Litwell's match modes build the right Boolean query for each source: All words is AND, Any word is OR, and Exclude terms act like NOT.

Citation count

The number of other works that cite a paper. A rough signal of influence; it differs between databases and favours older papers.

Crossref

A non-profit registry of journal articles and their DOIs. It covers every field but often lacks abstracts.

DOI

Digital Object Identifier: a permanent code for a paper, such as 10.1038/s41586-020-2649-2. Litwell uses it to recognise the same paper in different sources.

EPO

The European Patent Office. Its Open Patent Services give access to Espacenet, covering about 100 patent offices.

Europe PMC

A free biomedical literature database run by the European Bioinformatics Institute, covering PubMed content, preprints, patents and free full text.

Exact phrase

The topic is searched as one phrase, so 'B2 protein' does not match a paper that only has 'B2' and 'protein' far apart.

Exclude terms

A term in the title or journal removes the record; a term only in the abstract lowers its score.

Link

A page for the record (publisher, PubMed, Google Patents ...).

Must also contain

At least one of these terms must appear in the title or abstract, or the record is removed.

New Since Last Run

Yes when the record was not in this file before the latest run; No when it was. Blank for the first run into a new file.

Ollama

Free software that runs AI language models on your own computer.

Open access

Free to read online without a subscription. The Open-access link column points to a free copy when one is known.

Open-access link

A free full-text link, when a source reports one.

OpenAlex

A free, open index of papers, authors and institutions across all fields, with abstracts, citation counts and open-access links.

Preprint

A paper shared before peer review (bioRxiv, arXiv ...).

Profile

A named, saved set of search fields in Litwell.

PubMed

The US National Library of Medicine's index of biomedical literature.

Relevance Score

0 to 100, best first. Up to 50 points when your topic appears as a phrase in the title (otherwise up to 30 for the share of its words in the title); up to 25 when it is in the abstract (otherwise up to 15 for the share of words); 10 per 'Must also contain' term found, up to 20; 5 for a journal or field match; minus 20 per exclusion term found only in the abstract, up to 40. It is a rule-based guide, not a judgement of quality.

Search Date / Search Terms

When the record was first found and the fields used; a later search that finds it again adds its terms.

Search Log sheet

One row per source per run (including errors and skipped sources) and one row per filter rule with the number of rows it removed.

Summary sheet

Counts of what the file holds, by source, type, year, AI rating and score band.

Type

Paper, Preprint (not yet peer reviewed) or Patent.

USPTO

The United States Patent and Trademark Office. Its Open Data Portal lists granted US patents and published applications.